WHITEPAPER & PRACTICAL TUTORIAL

ARB Basket Fair Value [Approx]

How the Pine Script works, how to install it in TradingView, how to tune it, and how to interpret its signals

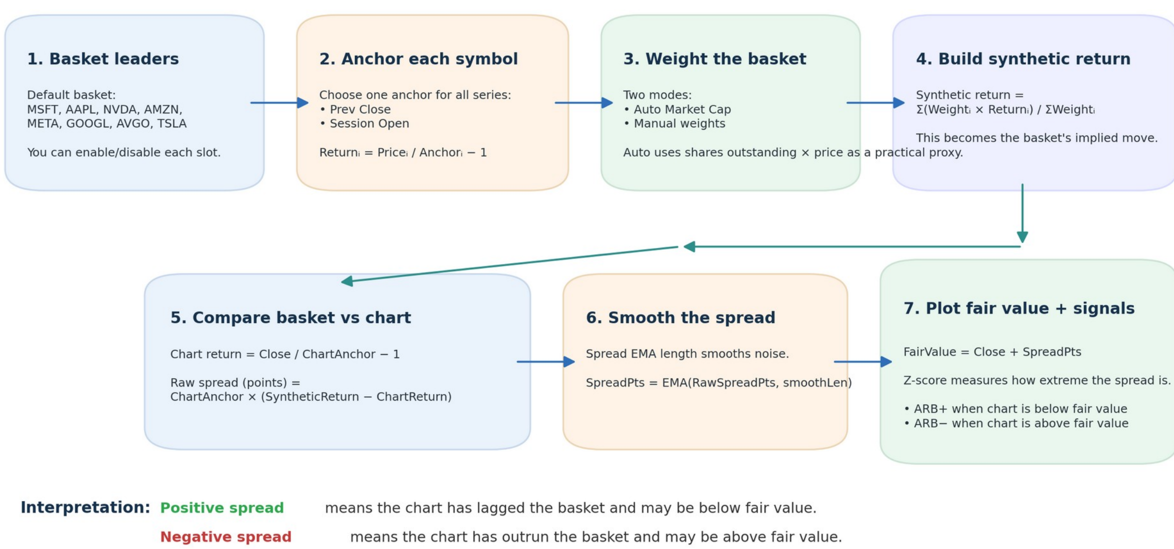
Prepared for: TradingView users of the ARB Basket Fair Value script • **Date:** April 16, 2026

Purpose. This document explains the public-logic Pine Script approximation that was built from publicly discussed arbitrage / fair-value ideas. It is designed to be practical: you can install the script, tune it, and use it without reverse-engineering the source every time.

> **Important note about scope**
> This script is not a verified 1:1 clone of any proprietary Rocket Scooter indicator. It is a transparent TradingView approximation built from public descriptions of basket-led, arbitrage-style fair-value logic. Treat it as a relative-value overlay and educational tool—not as a stand-alone trading system.

Figure 1. How the ARB Basket Fair Value script works

The script converts a basket of leaders into a synthetic fair-value estimate for the chart, then measures dislocation.



Concept overview of the script's data flow and decision logic.

Best suited for index ETFs and futures when the basket reflects the instrument's primary leaders.

Contents

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1. Executive summary

In one sentence: the script asks whether a chosen basket of leaders has moved more or less than the chart instrument since a common anchor, then translates that difference into a synthetic fair-value line and a z-score-based signal layer.

- A positive spread means the chart has lagged the basket and may be trading below basket-implied fair value.
- A negative spread means the chart has outrun the basket and may be trading above basket-implied fair value.
- The orange fair-value line is not a prediction of trend direction by itself; it is a relative-value estimate built from basket behavior.
- ARB+ and ARB− labels print only when the spread becomes statistically extreme relative to the selected z-score lookback.
- The summary table is the fastest way to inspect state: spread, z-score, basket return, chart return, and active-symbol count.

> **Best first use**
> Load the script on SPY or ES first, keep the default eight-symbol mega-cap basket, use Auto Market Cap weighting, keep Basket timeframe blank, and leave the threshold at 1.5 until you have seen how it behaves across several sessions.

Why this framework can be useful

Traditional price-only indicators lag by design. **This script tries to surface relative dislocation instead:** if leadership stocks have already moved, but the target chart has not fully followed, the spread can flag that gap before a simple moving-average crossover would.

2. What the indicator actually calculates

The script is straightforward once you break it into stages. Every basket symbol gets the same anchor logic, the same timeframe logic, and the same return formula. Those returns are then combined into one synthetic basket return, compared against the chart return, and transformed into points, fair value, and z-score.

Stage	Formula / logic	Interpretation
Anchor each symbol	Anchor_i = previous close OR session open	Defines the common baseline for all returns.

Compute symbol return	$Return_i = Price_i / Anchor_i - 1$	Measures how far each basket member moved from the anchor.
Assign weight	$Weight_i$ = Auto Market Cap OR manual input	Lets large leaders matter more, or equalizes them if you choose manual.
Aggregate basket	$SyntheticReturn = \Sigma(Weight_i \times Return_i) / \Sigma Weight_i$	Builds one combined basket move.
Compare to chart	ChartReturn = Close / ChartAnchor − 1	Measures the target chart from the same anchor.
Convert to points	RawSpreadPts = ChartAnchor × (SyntheticReturn − ChartReturn)	Expresses the dislocation in chart-price points.
Smooth	SpreadPts = EMA(RawSpreadPts, smoothLen)	Reduces noise and stabilizes the fair-value line.
Plot fair value	FairValue = Close + SpreadPts	Shows where the chart would sit if it matched basket behavior.
Standardize extremes	Z = (SpreadPts − mean) / stdev	Creates consistent signal thresholds across sessions.

Core Pine Script excerpt

```
float syntheticRet = active >= minActive and sumW > 0 ? sumWR / sumW : na
float rawSpreadPts = not na(syntheticRet) and not na(chartRet) and not na(chartAnchor) ?
chartAnchor * (syntheticRet - chartRet) : na
float spreadPts    = ta.ema(rawSpreadPts, smoothLen)
float fairValue    = not na(spreadPts) ? close + spreadPts : na

float spreadMean  = ta.sma(spreadPts, zLen)
float spreadStdev = ta.stdev(spreadPts, zLen)
float zScore      = not na(spreadPts) and not na(spreadStdev) and spreadStdev != 0 ? (spreadPts
- spreadMean) / spreadStdev : na
```

Translation: the basket defines the expected move; the chart is measured against it; the difference becomes the signal.

3. What the script is and is not

What it is

- A transparent Pine Script overlay that requests price and financial data for up to eight basket symbols.
- A basket-led fair-value model that compares synthetic basket behavior against the chart instrument.
- A practical approximation for ARB-style, arbitrage-themed dislocation analysis using public TradingView data.
- A readable base script that you can customize, extend, or adapt to different sectors and index proxies.

What it is not

- Not a verified clone of Rocket Scooter's internal calculations or proprietary data sources.
- Not an options-flow, dealer-positioning, or institutional-hedging model.
- Not float-adjusted S&P index math; Auto Market Cap uses a practical proxy based on shares outstanding × price.
- Not a complete trading plan. It still needs context, liquidity awareness, risk limits, and discretionary execution rules.

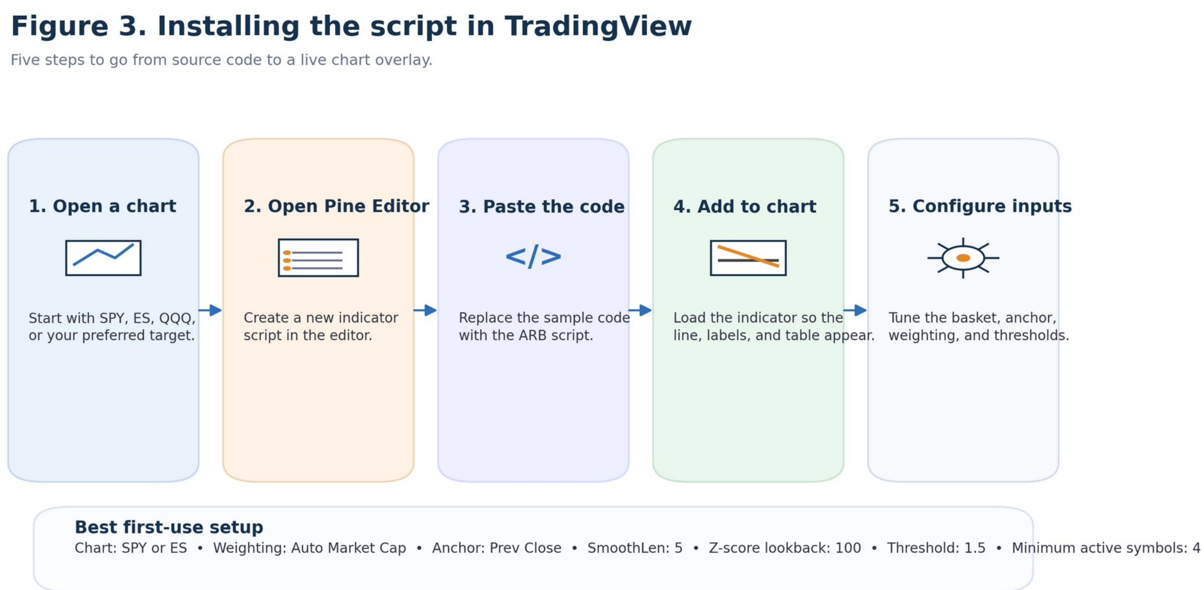
Why only eight basket slots?
The script makes multiple request.*() calls per symbol—price, anchor, and financial data. Keeping the basket to eight names is a practical tradeoff between interpretability, speed, and TradingView request limits. With the default structure, the script uses roughly 25 unique data requests.

Best mental model

Think of the script as a relative-value compass, not an oracle. It is strongest when the basket genuinely leads the instrument you are charting, and weakest when the basket has no real causal relationship to that chart.

4. Installation guide for TradingView

You only need a chart, the Pine Editor, and the script source. This section is written for speed: follow it once, save the indicator, then reuse it like any other custom TradingView tool.



From source code to live overlay in five steps.

Step-by-step

1. Open the chart you want to trade or analyze. Start with SPY, ES, QQQ, or NQ before testing more specialized instruments.
2. Open the Pine Editor and create a new indicator script.
3. Paste the ARB Basket Fair Value source code into the editor.
4. Save the script so it becomes part of your TradingView scripts list.
5. Click Add to chart and confirm that the fair-value line, labels, background shading, and summary table appear.
6. Open the indicator's Settings panel and review the General group first, then the Basket 1–8 groups.

Sanity check after installation
If the line does not appear, inspect the summary table first. If it says 'Insufficient basket,' your enabled symbols are missing data, unavailable on your plan, or below the minimum active-symbol count.

Where to save time

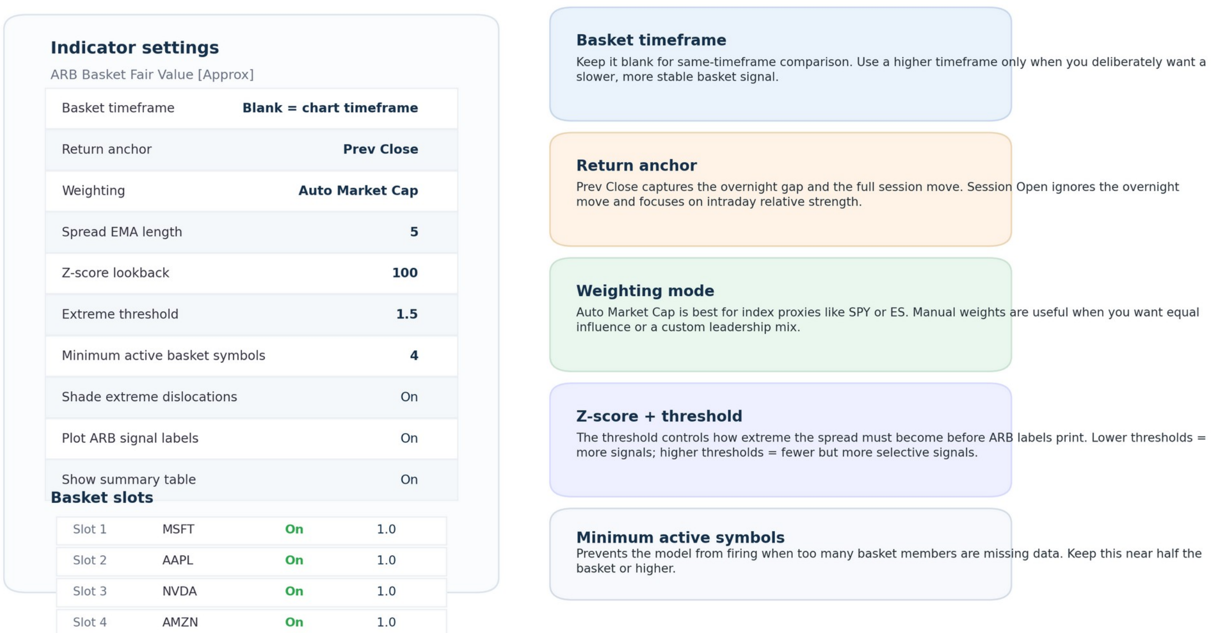
- Leave Basket timeframe blank on first use.
- Keep the default basket until you understand how the spread behaves.
- Do not optimize every setting on day one; watch how the line reacts for several sessions first.
- Create alerts only after you have validated that the basket and threshold match your instrument.

5. Settings and parameter reference

The script is intentionally configurable. The key is to understand which inputs change the model's logic versus which inputs only change presentation.

Figure 4. Settings guide: what each input does

Use the General group to control signal behavior, then customize the basket slots for your market.



General inputs govern the signal engine; basket slots determine what leads the model.

Input	Type	What it controls	Practical guidance
Basket timeframe	General	Whether the basket is read on chart timeframe or a specified timeframe.	Leave blank first. Use a higher timeframe only when you want a slower, more stable basket signal.
Return anchor	General	Common baseline for basket and chart returns.	Prev Close captures gap + session. Session Open isolates intraday behavior.
Weighting	General	How basket members influence synthetic return.	Auto Market Cap for index proxies; Manual for equal-weight or custom leadership.
Spread EMA length	General	Noise reduction applied to the spread.	Lower = faster and noisier. Higher = slower and smoother.
Z-score lookback	General	Window used to standardize spread extremes.	Short lookback adapts quickly; long lookback makes extremes harder to reach.
Extreme threshold	General	Signal sensitivity for ARB labels and background shading.	Lower threshold = more signals; higher threshold = more selectivity.
Minimum active basket symbols	General	Required symbol participation before model is considered valid.	Keep near half the basket or above.
ShowBg / ShowSignals / ShowTable	Display	Visual overlays and label visibility.	Turn off anything that distracts from execution.
Basket 1–8 slots	Basket	Which symbols participate and what manual weights they use.	Choose leaders that genuinely drive the target chart.

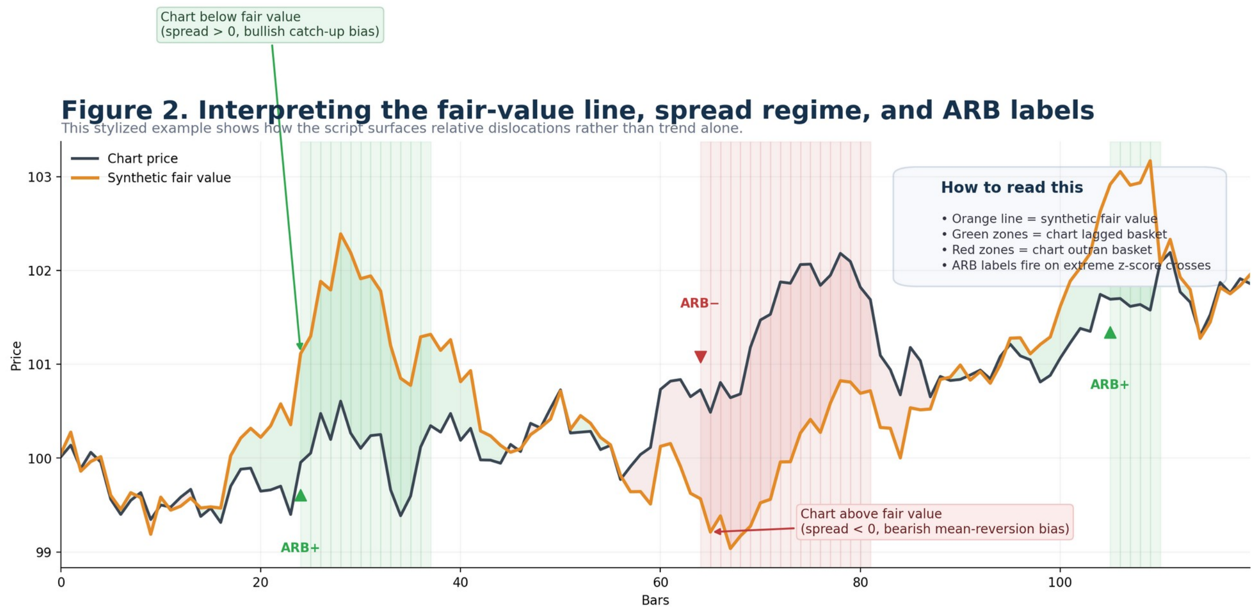
Fast tuning rules

- Too many labels? Increase threshold or increase smoothing.
- Signals feel late? Reduce smoothing or reduce lookback slightly.
- Fair-value line looks jumpy? Increase Spread EMA length.
- Model fails too often? Lower minimum active symbols only if you understand why data is missing.
- Basket feels wrong? Fix the basket before touching thresholds.

6. Reading the chart, labels, and summary table

A. Reading the fair-value line and shading

The plotted fair-value line answers one question: **where would the chart be if it matched the basket's weighted move?** When the line sits above price, the chart is lagging. When it sits below price, the chart is leading.

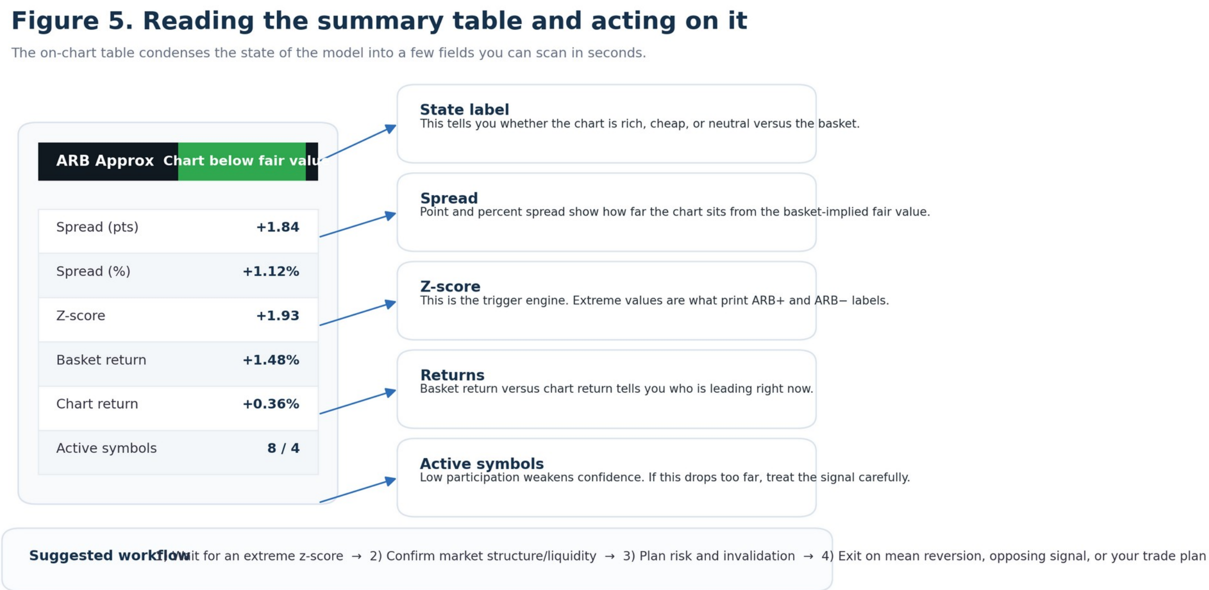


Stylized example of chart price versus synthetic fair value.

Chart condition	What the script sees	Visual cue	Interpretation
Fair value above price	Basket stronger than chart	Green context / ARB+ when extreme	The chart may be below basket-implied fair value.
Fair value below price	Chart stronger than basket	Red context / ARB− when extreme	The chart may be above basket-implied fair value.
Z-score near zero	Spread normalized	No extreme state	Relative dislocation has mostly mean-reverted.
Insufficient basket	Too few valid symbols	Neutral / gray state text	Model confidence is reduced; fix data or basket design.

B. Reading the summary table

You can think of the on-chart table as the model dashboard. If you only have two seconds to check the indicator, read the state label, the z-score, and the active-symbol count.



The summary table is the fastest diagnostic view during live market use.

Practical reading order
1) Is the basket valid? 2) Is the state rich, cheap, or neutral? 3) Is the z-score extreme enough to matter? 4) Is price action actually responding? 5) Does the trade still make sense once you apply risk, news, and liquidity context?

7. Suggested presets and basket design

The basket matters more than any single threshold. If the basket does not genuinely lead the chart, the spread becomes noise. Start by matching your chart to the names that most directly drive it.

Target chart	Suggested basket	Anchor	Start settings	Why it works
SPY / ES	MSFT, AAPL, NVDA, AMZN, META, GOOGL, AVGO, TSLA	Prev Close	EMA 5 | Z 100 | Thr 1.5	Large index proxies often respond to the same mega-cap leadership captured by the default basket.
QQQ / NQ	AAPL, MSFT, NVDA, AMZN, META, GOOGL, AVGO, TSLA	Prev Close	EMA 5 | Z 100 | Thr 1.5	The default basket is already tech-heavy and aligns naturally with Nasdaq-style leadership.
SMH / SOX-style charts	NVDA, AVGO, AMD, QCOM, MU, ASML, TSM, TXN	Prev Close	EMA 5 | Z 80 | Thr 1.6	A semiconductor-specific basket reduces unrelated noise.
XLK / Tech sector	AAPL, MSFT, NVDA, AVGO, ORCL, CRM, ADBE, AMD	Prev Close	EMA 5 | Z 100 | Thr 1.6	Useful when you want sector leadership rather than the broader index.
Any instrument with custom leaders	Your own 4–8 symbols	Prev Close or Session Open	Start conservative, then observe	Manual basket design is often better than over-optimizing the threshold.

When to use Session Open instead of Prev Close

- Use Prev Close when overnight gap behavior is part of the story you want the model to capture.

- Use Session Open when you want to isolate only the intraday move from the opening bell onward.
- If your market has highly distorted overnight action, Session Open can make the relative-value read cleaner.

Manual weight design ideas

- Equal weights when you want every basket member to have the same influence.
- Heavier weights on the most causally important leaders when one or two names dominate the sector.
- Lighter weights on symbols that are volatile but less relevant to your target chart.
- Avoid stuffing the basket with loosely related names just because they are liquid or popular.

Simple rule for basket selection
Choose symbols that would make you say, 'If these move first, my chart usually reacts next.' If you cannot say that with confidence, redesign the basket before you trust the indicator.

8. Alerts, workflow, and trade planning

Built-in alert conditions

```
alertcondition(ta.crossover(zScore, zThresh), "ARB long dislocation", "Chart moved below
synthetic basket fair value.")
alertcondition(ta.crossunder(zScore, -zThresh), "ARB short dislocation", "Chart moved above
synthetic basket fair value.")
alertcondition(ta.cross(zScore, 0), "ARB mean reversion", "ARB spread crossed back through
zero/mean area.")
```

How to use them: open TradingView's Alerts dialog, choose this indicator as the condition source, then select one of the three built-in conditions. The long and short dislocation alerts are the entry-style events; the mean-reversion alert is the normalization event.

A practical discretionary workflow

Workflow step	What to do
1. Validate the basket	Confirm that the active-symbol count is healthy and the basket really belongs with the chart.
2. Wait for extreme state	Do not react to every wobble. Let the z-score reach your threshold or higher.
3. Confirm context	Check liquidity, news, opening auction behavior, and obvious market structure before acting.
4. Define risk first	Decide the invalidation level before you enter. The indicator gives context; it does not size risk for you.
5. Manage the trade	Targets can be fair value, zero-cross, opposing signal, or your preplanned structural level.
6. Review afterward	Keep notes on which baskets and thresholds produce stable versus noisy behavior.

Examples of reasonable uses

- Fade a stretched move when the chart is materially above fair value and the spread is at a negative extreme.
- Look for catch-up continuation when the chart is below fair value but begins reclaiming structure as the basket remains strong.
- Use the mean-reversion alert as a prompt to reduce exposure or reassess whether the relative-value edge has already normalized.

Risk note
A relative-value signal can stay extreme longer than expected, especially around macro releases, earnings, or forced flows. Never assume the first ARB label is automatically the best entry. Wait for execution context.

9. Troubleshooting, limitations, and extension ideas

Symptom	Likely cause	What to try
'Insufficient basket' state	Too many missing or invalid symbols; minActive too high; symbol data unavailable.	Check each enabled basket symbol, reduce the basket to known-available names, or lower minActive only if justified.
Too many ARB labels	Threshold too low or smoothing too fast for the chart timeframe.	Increase threshold and/or Spread EMA length.
Signals feel too slow	Smoothing too high or z-score lookback too long.	Reduce smoothing or shorten z-score lookback modestly.
Fair-value line seems unrelated	Basket does not actually lead the target chart.	Redesign the basket before touching any other setting.
Auto Market Cap behaves oddly on non-stocks	Financial data may be unavailable or inconsistent for some symbols.	Use Manual weighting for ETFs, futures proxies, or symbols without usable financial fields.
Performance / request issues	Too many requested datasets for plan or symbol mix.	Keep the basket moderate and avoid unnecessary expansion of request-heavy logic.

Known limitations

- Auto Market Cap is a proxy, not true float-adjusted index weighting.
- The script uses public chart + financial data only; it does not ingest options positioning, gamma, dealer hedging, or proprietary flows.
- A fair-value estimate can diverge for long stretches when macro or event risk dominates relative-value behavior.
- Signals on very low-liquidity charts are less trustworthy because spread mechanics can reflect noise rather than genuine leadership.

Easy extensions if you want to modify the source

- Add more basket presets by changing the input.symbol() defaults.
- Switch the default threshold or smoothing values for your preferred timeframe.
- Expose z-score itself as a separate plot in a lower panel by changing overlay = false and plotting zScore.
- Replace static basket blocks with arrays and loops if you want a more compact codebase.
- Add entry filters such as trend confirmation, session filters, or minimum ATR conditions.

Where to edit the source quickly

Customization target	Where to edit
Change indicator name	indicator("ARB Basket Fair Value [Approx]", ...)
Change default basket symbols	Each input.symbol(...) line inside Basket 1–8 groups
Change default threshold	input.float(1.5, "Extreme threshold (z-score)", ...)
Change summary table position	table.new(position.top_right, 2, 7, ...)
Change alert wording	Each alertcondition(...) line
Change line color logic	fvColor = underValued ? ... : overValued ? ... : ...

10. One-page quick reference

Question	Fast answer
What does ARB+ mean?	The chart is below basket-implied fair value by an extreme positive z-score.
What does ARB− mean?	The chart is above basket-implied fair value by an extreme negative z-score.
Best first chart?	SPY or ES with the default basket and Auto Market Cap.
Best first anchor?	Prev Close.
What if the line looks noisy?	Increase Spread EMA length.
What if there are too many signals?	Raise the threshold or extend lookback modestly.
What if Auto Market Cap fails?	Use Manual weighting.
What should I read first in the table?	State label, z-score, and active symbols.
What matters most?	A basket that truly leads the chart.

Shortest useful checklist

Match the basket to the chart. Validate symbol participation. Wait for an extreme spread. Confirm context. Define risk first. Then decide whether the fair-value gap is worth trading.

11. References

The script and this tutorial were informed by public documentation and public-facing descriptions only. The sources below are included so you can verify the model assumptions and platform constraints for yourself.

7. TradingView User Manual. Other timeframes and data. Covers request.security() and request.financial(), including cross-symbol and financial-data requests.
8. TradingView Help Center. I see the 'The script executes too many unique request.*() function calls' error. Useful for understanding request limits and why basket size matters.
9. AP Research (AlphaPicks). An Interview With RocketScooter CEO, Matt Cowart. Public discussion of forward-looking indicators, arbitrage mechanics, and market-cap-weighted aggregate behavior.
10. RocketScooterAI YouTube Shorts page. Public short-form posts including 'Our Leading Indicators Vs Lagging Indicators.'
11. RocketScooterAI Instagram reel title: 'The World's First Leading Indicator - This is what the ARB is.'

For convenience, keep the Pine Script source alongside this document so you can cross-reference the sections above with the actual code.